Literature Reviews and Meta-Analyses

From Research Questions to Evidence Synthesis

Nils Handler

University of Zurich (UZH)

October 8, 2026

The Evidence Gap

Why rigorous evaluation is missing in climate policy

Stechemesser et al. 2024

4.2%

of 1'500 policies implemented that could be attributed to a clean break in the emissions trajectory of a country.

EFI Review (2024)

73%

of German RI evaluations claimed causal effects, but only 24% actually used experimental or quasi-experimental designs.

Key Insight

Policymakers are making billion-dollar decisions based on weak evidence. Systematic reviews aggregate rigorous studies to create a stronger evidence base.

Defining Your Research Question

The PICO Framework (outline Nishikawa-Pacher 2022)

P - Population: Who are the study subjects? Be specific about geography/type. *Example: Households in OECD countries*
I - Intervention: What treatment or policy is being evaluated? *Example: Monetary incentives (rebates)*
C - Comparison: What is the control or alternative approach? *Example: Behavioral nudges*
O - Outcome: What measurable, quantifiable result is used? *Example: Energy consumption (kWh)*

Real World Application: Khanna et al. 2021

"For households in 25 countries (P), how effective are monetary incentives (I) compared to non-monetary nudges (C) in reducing residential energy consumption (O)?"

Types of Literature Reviews

From subjective narratives to rigorous synthesis (LitMaps 2026)

Narrative Review: Subjective synthesis of selected studies. Good for background, but prone to bias.
Systematic Review: Answers a specific question using a pre-defined, reproducible protocol.
Meta-Analysis: Statistically combines effect sizes from multiple studies to find an average effect.
Scoping Review: Maps the landscape of research on a broad concept connected to a population to identify gaps.

Key Distinction

Systematic reviews are transparent (anyone can reproduce them), while narrative reviews are subjective (depend on the reviewer's choices).

New Developments in Literature Reviews

Evidence Gap Map: Visualizes where evidence exists and where it is missing for a specific intervention space. (Example: Saran and White 2018)

Living Literature Review: A repository which contains a summary of the literature on a subject which is version controlled and updated with the literature. Avoids becoming outdated quickly and repeated work. (Example: VoxDev 2026)

Living Meta-Analysis: Same as the living literature review, but for a meta-analysis which mathematically evaluates and combines effects from different studies. (Example: Iversen et al. 2026)

The PRISMA Framework

Preferred Reporting Items for Systematic reviews and Meta-Analyses

Identification: How did you search? Document databases, keywords, and date ranges.
Screening: Filter titles and abstracts. Apply inclusion/exclusion criteria.
Eligibility: Full-text review. Assess quality and risk of bias.
Inclusion: Final sample for data extraction and synthesis.

Key Goal

Reproducibility. Another researcher using your protocol should get the exact same results.

Search Strategy & Study Selection

How to find studies without bias

- **Databases**: Where did you search? (e.g., PubMed, Web of Science, JSTOR, Google Scholar)
- **Keywords**: What search strings did you use? (e.g., "energy consumption" AND "behavioral intervention")
- **Date Range**: What time period? (e.g., 2010-2024)
- **Inclusion Criteria**: What qualifies? (e.g., RCTs only, quantitative outcomes)
- **Exclusion Criteria**: What is rejected? (e.g., no control group, non-peer-reviewed, wrong sector)

Real Example: Stechemesser et al. 2024

Searched the OECD policy database for 1,500 climate policies across 41 countries (1998-2022).

Data Extraction & Standardization

Converting disparate outcomes into comparable metrics

The Challenge: Heterogeneity

Studies report outcomes in different units (e.g., %, kWh, €, CO), preventing direct comparison.

The Solution: Standardization

1. Effect Sizes
2. Relative Change (%)
3. Absolute Conversion

Real World Example: Khanna et al. 2021

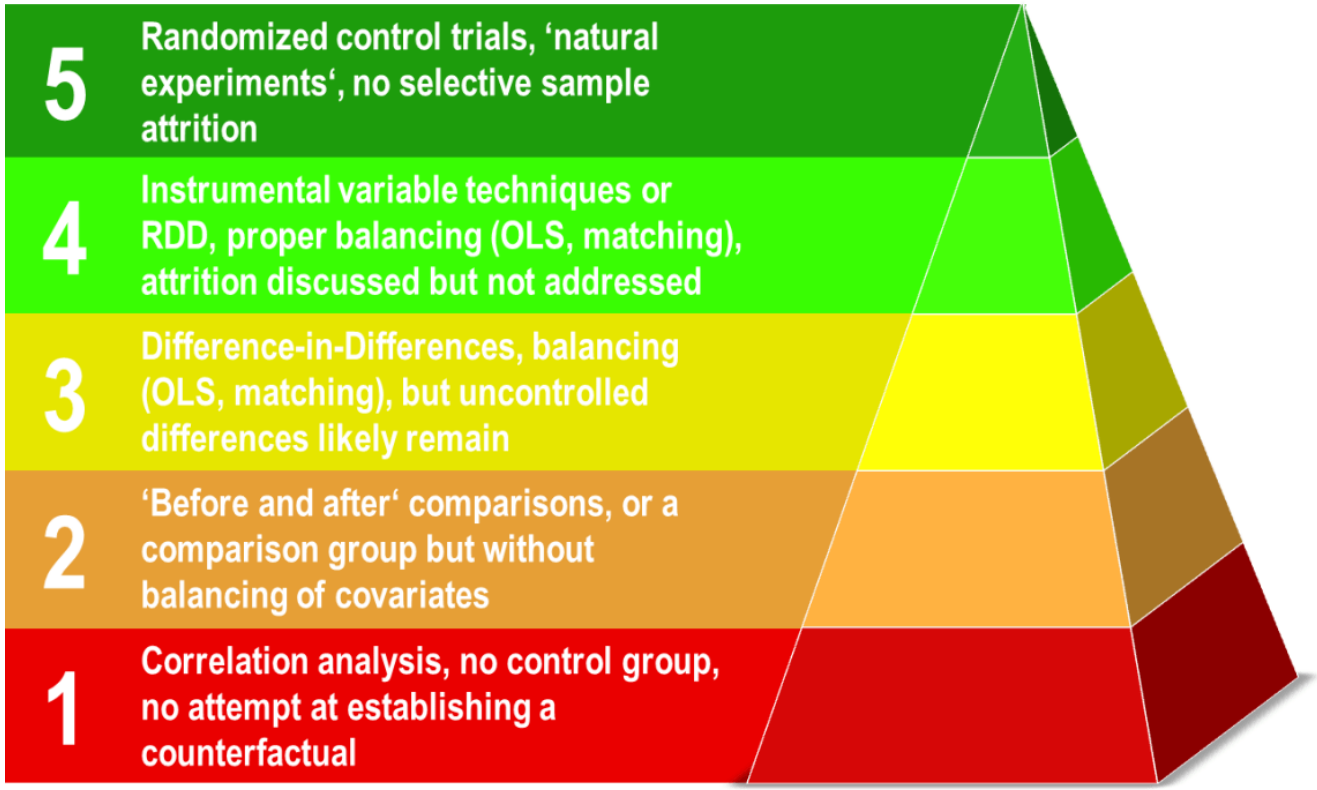
From 122 studies using various different units, they extracted 360 effect sizes and converted them to Fisher's Z, enabling statistical comparison.

Quality Assessment & Risk of Bias

Not all studies are created equal

Many different fields use hierarchies of evidence to compare different studies types. While there are exceptions, they are a useful tool for helping classify evidence.

Scientific Maryland Scale (SMS) Hierarchy of Evidence



Risk of Bias

- **Selection Bias**: Units select into different treatment groups based on a variable actually impacting the outcome of interest (i.e. units want to receive a new treatment).
- **Attrition Bias**: Units from different groups or treatments leave the study in different rates (i.e. people find out they are in a placebo group).
- **Reporting Bias**: Particularly a risk when self-reporting, units inaccurately report their results along a consistent logic (i.e. under-reporting of income).

For a comprehensive list of bias types, see the Catalog of Bias (David Nunan, Carl Heneghan, and David Sackett 2026).

Meta-Analysis Basics

Meta-Analysis vs Literature Review

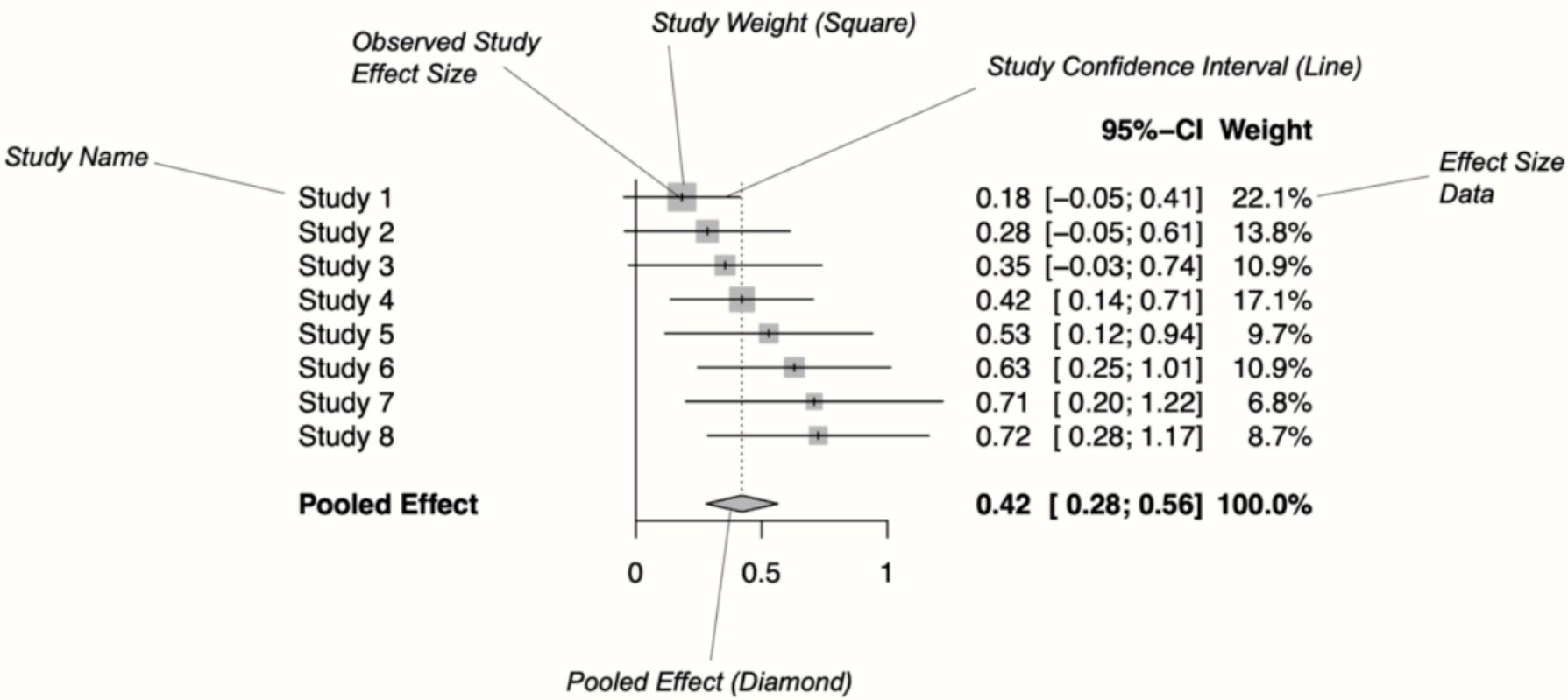
A meta-analysis is a type of literature review where evidence is statistically combined across multiple studies. Literature reviews do not need to have this quantitative analysis.

Key ideas to keep in mind

- **Effect Size**: Standard test statistics that can be compared across papers, typically unitless (e.g., Odds Ratio, Cohen's d).
- **Weighting (Inverse Variance)**: Larger, more precise studies are given more weight.
- **Heterogeneity**: Do studies disagree more than expected by chance?

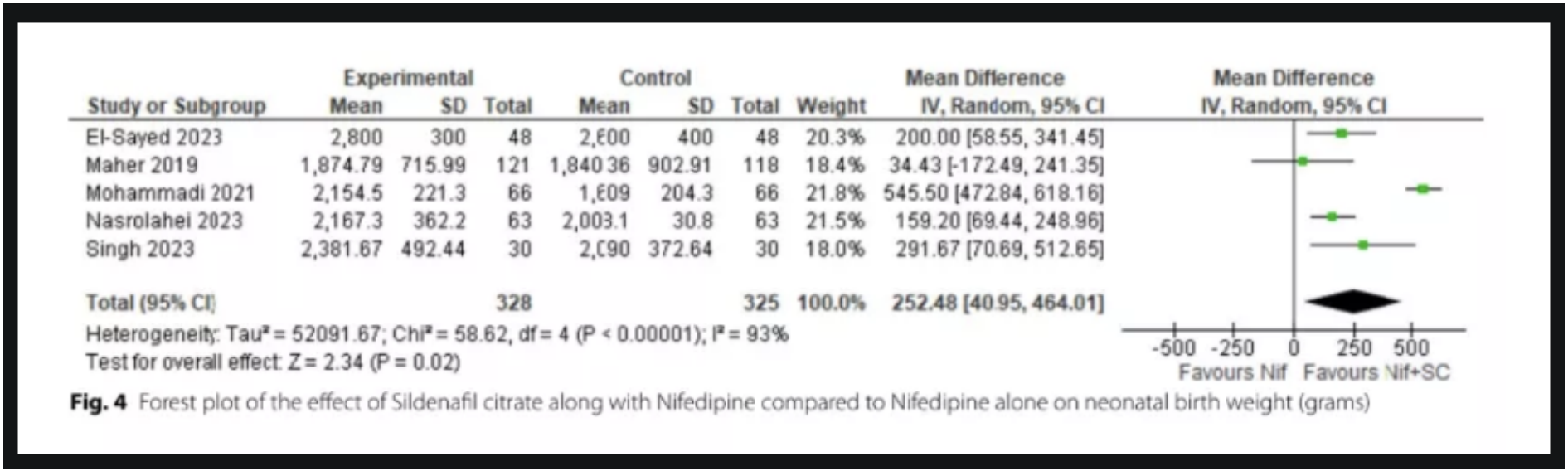
The Forest Plot

The standard visualization for meta-analysis. All studies and their confidence intervals are compared.



Forest Plots

A tool for flagging suspect studies!



Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Random, 95% CI
El-Sayed 2023	2,800	300	48	2,600	400	48	20.3%	200.00 [58.55, 341.45]
Maher 2019	1,874.79	715.99	121	1,840.36	902.91	118	18.4%	34.43 [-172.49, 241.35]
Mohammadi 2021	2,154.5	221.3	66	1,609	204.3	66	21.8%	545.50 [472.84, 618.16]
Nasrolahei 2023	2,167.3	362.2	63	2,003.1	30.8	63	21.5%	159.20 [69.44, 248.96]
Singh 2023	2,381.67	492.44	30	2,090	372.64	30	18.0%	291.67 [70.69, 512.65]
Total (95% CI)			328			325	100.0%	252.48 [40.95, 464.01]

Heterogeneity: $Tau^2 = 52091.67$; $Chi^2 = 58.62$, $df = 4$ ($P < 0.00001$); $I^2 = 93\%$

Test for overall effect: $Z = 2.34$ ($P = 0.02$)

Fig. 4 Forest plot of the effect of Sildenafil citrate along with Nifedipine compared to Nifedipine alone on neonatal birth weight (grams)

Does anything jump out at you when looking at this forest plot?

Machine Learning in Literature Reviews

Scaling systematic reviews to big data

Automated Screening

NLP algorithms can classify thousands of documents instantly.

Break Detection

Algorithms identify structural breaks in time-series data.

Variable Selection

Lasso regression and Indicator Saturation select relevant policy drivers.

Case Study: Stechemesser et al. 2024

They analyzed 1,500 climate policies across 41 countries. Using ML break detection, they identified 69 major emission reductions and matched them to 63 specific policies (or policy mixes).

Causal Inference in Reviews

Synthesizing evidence, not just correlations

The Trap: Naive Synthesis

- Vote Counting
- Ignoring Selection Bias

The Solution: Causal Filtering

- Filter by Design
- Quality Weighting

Critically Screen Studies

Garbage In, Garbage Out. A meta-analysis cannot fix a biased original study.

Disentangling Effects

Isolating the impact of interacting interventions

- The "Cocktail" Problem: Policies rarely happen in isolation.
- **Temporal Variation (Timing)**: Exploit staggered implementation dates.
- **Spatial Variation (Geography)**: Compare regions with different policy bundles.
- **Interaction Analysis**: Explicitly model the interaction term (Policy A $\times$ Policy B).

Key Finding: Stechemesser et al. 2024

Synergies matter: Subsidies alone often failed to reduce emissions, but became highly effective when combined with other policies.

Publication Bias & Heterogeneity

Interpreting the silence and the noise

Publication Bias

The "File Drawer Problem".
Detection Tool: Funnel Plot.

Heterogeneity

When studies differ significantly (High I^2), investigate why with Subgroup Analysis.

Extreme Example: EFI Review (2024)

59 of 59 German R&I evaluations produced by German ministries (Research and Economics and Climate Action) using causal methods reported positive effects of the policy evaluated.

Funnel Plots

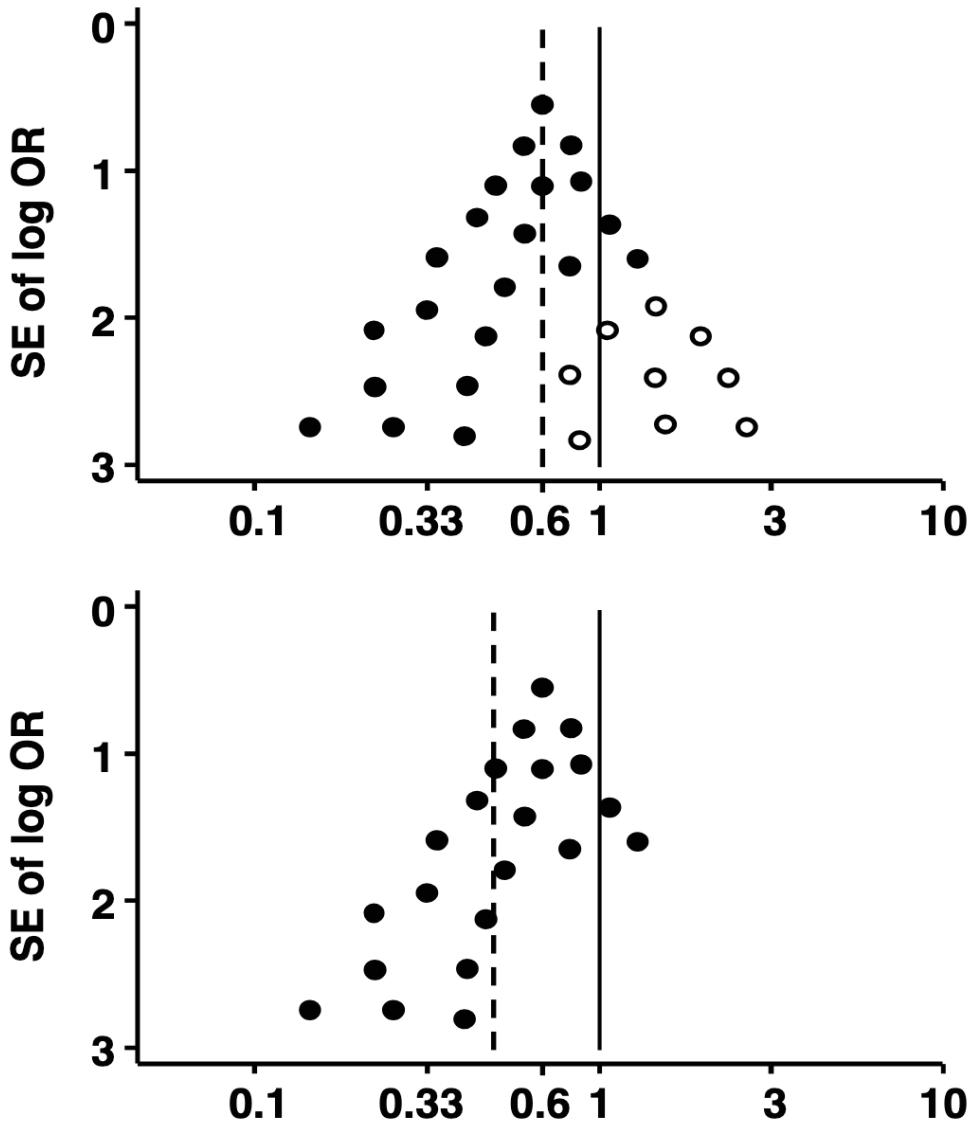
A tool for identifying publication bias

Funnel Plot

Horizontal Axis: Estimated effect size, typically an odds ratio, from different studies

Vertical Axis: Precision (inverse of variance) of the study

Image from Sterne and Harbord 2004



Dealing with Heterogeneity

Identifying and explaining why studies differ from one another

Heterogeneity: Identifying and quantifying how studies vary within your analysis. Key concepts to look consider Harrer et al. 2026a

- Two main types of heterogeneity: Design-related and statistical. Design-related are dealt with filtering out using PICO
- Quantifying: Calculate how much sampling error explain the difference between studies using common statistics (i.e. Cochran's Q or Higgins & Thompson's I^2)
- Identifying influential cases: Find outlier studies and perform checks to see how much they are impacting your meta-analysis estimates

Subgroup Analysis: With identified different studies, we check how they vary by comparing the subgroup's heterogeneity statistics to that of the group, then comparisons of hypotheses for explanations are made.

Warning!

These methods can have low power. Harrer et al. 2026c

Reporting & Transparency

Ensuring reproducibility and trust

- **Pre-Registration**: Define hypotheses and analysis plan before looking at data (OSF, AEA RCT Registry).
- **Open Data & Code**: Share raw data and analysis code.
- **Reporting Standards**: Use standardized checklists (PRISMA, CONSORT, TREND).

Necessary in a literature review!

Without these three pillars, a literature review is just a collection of anecdotes, not scientific evidence.

Case Study: Khanna et al. 2021

A blueprint for a rigorous meta-analysis

1. **Thorough Search**: Used 28 keyword combos across databases.
2. **Standardization**: Converted diverse measures into Fisher's Z.
3. **Subgroup Analysis**: Broken down effects by incentive type.
4. **Quality Control**: Tested for publication bias and isolated high-quality studies.

The Takeaway

A clear template for moving from reading papers to reproducible, quantitative synthesis.

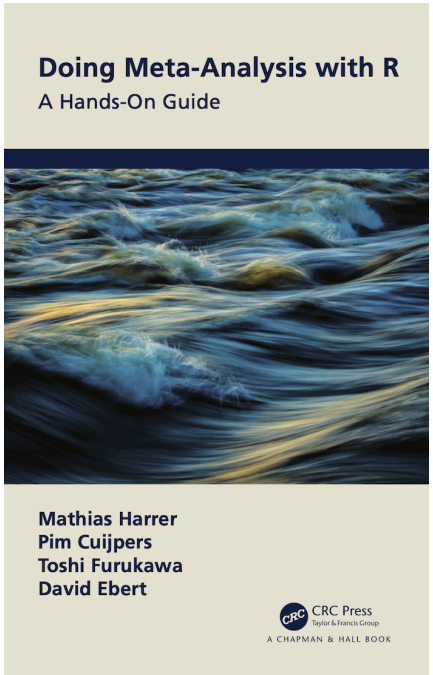
Best Resources to Use

For conducting the analysis:

A full, robust guide to most of the tools above and with instructions in *R*.

Harrer, M., Cuijpers, P., Furukawa, T.A., Ebert, D.D. (2021). Doing Meta-Analysis with R: A Hands-On Guide. Boca Raton, FL and London: Chapman Hall/CRC Press. ISBN 978-0-367-61007-4.

https://doing-meta.guide/



For finding papers in the first place:

- Connected Papers provides a visual overview.
- Semantic Scholar for a more traditional academic search engine.

Course Projects To-Dos

Some suggestions for where to start working

No lecture for the next two weeks.

What should you do in this time?

1. Review the materials available on OLAT for project ideas (and good references for learning more in-depth on class topics).
2. Finalize your project idea and inform us via email before the next class. Include the project type.

Course Projects To-Dos

Some suggestions for where to start working

For experimental design projects

3. Outline the question you want to understand and the intervention/treatment this entails.
4. Review the literature for similar studies for both inspiration and estimate for power (to be discussed in-depth later).
5. Identify what you are comparing it to and how you want to measure that gap.
6. List variables which could impact your outcome metric

For literature review projects

3. Outline the question you want to understand using the PICO format.
4. Check for pre-existing literature reviews to see what is already understood at the meta-level.
5. Outline your search criteria using the PRISMA Framework, including how you might exclude papers.

Course Projects To-Dos

Don't forget to come to the next class with questions about what you feel you need help on!

Citations and Resources I

- David Nunan, Carl Heneghan, and David Sackett (2026). *Biases | Catalog of Bias*. en. URL: https://catalogofbias.org/biases/ (visited on 02/26/2026).
- *Gutachten zu Forschung, Innovation und Technologischer Leistungsfähigkeit Deutschlands 2024* (2024). en. Tech. rep. Commision of Experts for Research and Innovation.
- Harrer, Mathias et al. (2026a). *Chapter 3 Effect Sizes | Doing Meta-Analysis in R*. en. URL: https://doing-meta.guide/effects/ (visited on 03/02/2026).
- — (2026b). *Chapter 5 Between-Study Heterogeneity | Doing Meta-Analysis in R*. en. URL: https://doing-meta.guide/heterogeneity/ (visited on 03/02/2026).

Citations and Resources II

- Harrer, Mathias et al. (2026c). *Chapter 7 Subgroup Analyses | Doing Meta-Analysis in R*. en. URL: https://doing-meta.guide/subgroup/ (visited on 03/03/2026).
- Hünermund, Paul (May 2016). *Does Innovation Policy Really Work?* en. URL: https://p-hunermund.com/2016/05/30/does-innovation-policy-really-work/ (visited on 02/26/2026).
- Iversen, Ingebjørg A. et al. (Mar. 2026). "The active monitoring of oxytocin research evidence (AMORE) platform". In: *Psychoneuroendocrinology* 185, p. 107713. ISSN: 0306-4530. DOI: 10.1016/j.psyneuen.2025.107713. URL: https://www.sciencedirect.com/science/article/pii/S0306453025004366 (visited on 02/26/2026).
- James Heathers (Feb. 2025). "An Introduction To Forensic Metascience". en. In: DOI: 10.5281/zenodo.14871843. URL: https://jamesheathers.curve.space/ (visited on 03/02/2026).

Citations and Resources III

- Khanna, Tarun M. et al. (July 2021). "A multi-country meta-analysis on the role of behavioural change in reducing energy consumption and CO2 emissions in residential buildings". en. In: *Nature Energy* 6.9, pp. 925–932. ISSN: 2058-7546. DOI: 10.1038/s41560-021-00866-x. URL: https://www.nature.com/articles/s41560-021-00866-x (visited on 02/26/2026).
- LitMaps (2026). *What are the different types of literature review?* en. URL: https://www.litmaps.com/learn/what-are-the-different-types-of-literature-review (visited on 02/26/2026).
- Nishikawa-Pacher, Andreas (June 2022). "Research Questions with PICO: A Universal Mnemonic". en. In: *Publications* 10.3. ISSN: 2304-6775. DOI: 10.3390/publications10030021. URL: https://www.mdpi.com/2304-6775/10/3/21 (visited on 02/26/2026).

Citations and Resources IV

- PRISMA Executive (2026). *PRISMA statement*. en-US. URL: https://www.prisma-statement.org (visited on 02/26/2026).
- Saran, Ashrita and Howard White (Oct. 2018). "Evidence and gap maps: a comparison of different approaches". In: *Campbell Systematic Reviews* 14.1, pp. 1–38. ISSN: 1891-1803. DOI: 10.4073/cmdp.2018.2. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC8428058/ (visited on 02/26/2026).
- Stechemesser, Annika et al. (2024). "Climate policies that achieved major emission reductions: Global evidence from two decades". In: *Science*. DOI: 10.1126/science.adl6547. URL: https://www.science.org/doi/10.1126/science.adl6547.

Citations and Resources V


Sterne, Jonathan A. C. and Roger M. Harbord (June 2004). "Funnel Plots in Meta-analysis". en. In: *The Stata Journal: Promoting communications on statistics and Stata* 4.2, pp. 127–141. ISSN: 1536-867X, 1536-8734. DOI: 10.1177/1536867X0400400204. URL: https://journals.sagepub.com/doi/10.1177/1536867X0400400204 (visited on 03/03/2026).


VoxDev (2026). *All VoxDevLit*. en. URL: https://voxdev.org/voxdevlit (visited on 02/26/2026).